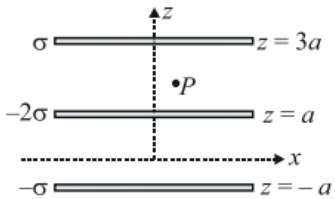
Lakshya JEE 2.0 (2025)

Physics

DPP: 4

Electric Charges and Fields

Q1 Three infinitely long charge sheets are placed as shown in figure. The electric field at point P is



(A) $\frac{2\sigma}{\varepsilon_0}\hat{k}$.
(B) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$
(C) $\frac{4\sigma}{\varepsilon_0}\hat{k}$
(D) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$

Q2 A semi-circular arc of radius a is charged uniformly and the charge per unit length is λ. the electric field at its centre is
(A) $\frac{\lambda}{2\pi\varepsilon_0 a^2}$
(B) $\frac{\lambda}{2\pi\varepsilon_0 a}$
(C) $\frac{\lambda^2}{\pi\varepsilon_0 a}$
(D) $\frac{\lambda}{4\pi\varepsilon_0 a}$

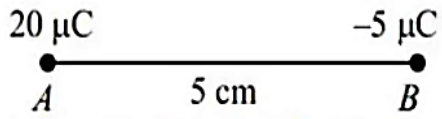
Q3 The magnitude of electric field at distance r from an infinitely thin rod having a linear charge density λ is (use Gauss's law)
(A) $E=\frac{\lambda}{2\pi\varepsilon_0 r}$
(B) $E=\frac{2\lambda}{\pi\varepsilon_0 r}$
(C) $E=\frac{\lambda}{4\pi\varepsilon_0 r}$
(D) $E=\frac{4\lambda}{\pi\varepsilon_0 r}$

Q4 If the linear charge density of an infinitely long cylinder $(r=2\text{ cm})$ is $4\mu\text{Cm}^{-1}$ then electric field intensity at point 3.6 cm from axis is
(A) $4\times10^5\text{NC}^{-1}$
(B) $2\times10^6\text{NC}^{-1}$
(C) $8\times10^7\text{NC}^{-1}$
(D) $12\times10^7\text{NC}^{-1}$

Q5 What is the field at the centre of ring of radius R carrying a charge q uniformly distributed over it?
(A) $\frac{Kq}{R}$
(B) 0
(C) $\frac{Kq}{2R}$
(D) $\frac{2Kq}{R}$

Q6 The maximum electric field intensity on the axis of a uniformly charged ring of charge q and radius R will be:
(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{3\sqrt{3}R^2}$
(B) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3R^2}$
(C) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{3\sqrt{3}R^2}$
(D) $\frac{1}{4\pi\varepsilon_0}\frac{3q}{2\sqrt{3}R^2}$

Q7 Two particles A and B having charges $20\mu\text{C}$ and $-5\mu\text{C}$ respectively are held with a separation of 5 cm. At what position a third charged particle should be placed so that it does not experience a net electric force?



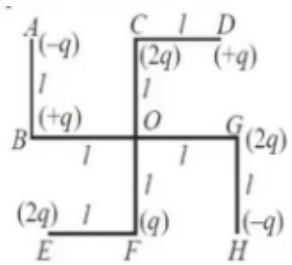
(A) At 5 cm from $20\mu\text{C}$ on the left side of system

(B) At $5\ \text{cm}$ from $-5\mu\text{C}$ on the right side

(C) At $1.25\ \text{cm}$ from $-5\mu\text{C}$ between two charges

(D) At midpoint between two charges

Q8 What will be the magnitude of electric field at point O as shown in the figure? Each side of the figure is l and perpendicular to each other?



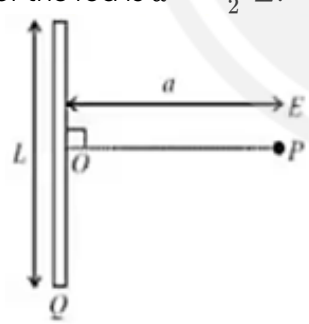
(A) $\frac{1}{4\pi\varepsilon_0}\frac{q}{l^2}$

(B) $\frac{1}{4\pi\varepsilon_0}\frac{q}{(2l^2)}(2\sqrt{2-1})$

(C) $\frac{q}{4\pi\varepsilon_0(2l^2)}$

(D) $\frac{1}{4\pi\varepsilon_0}\frac{2q}{(2l^2)}(\sqrt{2})$

Q9 Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the rod is $a = \frac{\sqrt{3}}{2}L$.



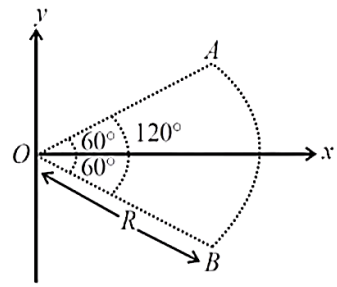
(A) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$

(B) $\frac{Q}{3\pi\varepsilon_0 L^2}$

(C) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$

(D) $\frac{Q}{4\pi\varepsilon_0 L^2}$

Q10 Figure shows a rod AB, which is bent in a 120° circular arc of radius R. A charge $(-Q)$ is uniformly distributed over rod AB. What is the electric field E at the centre of curvature O ?



(A) $\frac{3\sqrt{3}Q}{8\pi\varepsilon_0 R^2}(\hat{i})$

(B) $\frac{3\sqrt{3}Q}{8\pi^2\varepsilon_0 R^2}(\hat{i})$

(C) $\frac{3\sqrt{3}Q}{16\pi^2\varepsilon_0 R^2}(\hat{i})$

(D) $\frac{3\sqrt{3}Q}{6\pi^2\varepsilon_0 R^2}(\hat{i})$

Answer Key

Q1	(B)	Q6	(C)
Q2	(B)	Q7	(B)
Q3	(A)	Q8	(B)
Q4	(B)	Q9	(C)
Q5	(B)	Q10	(B)

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